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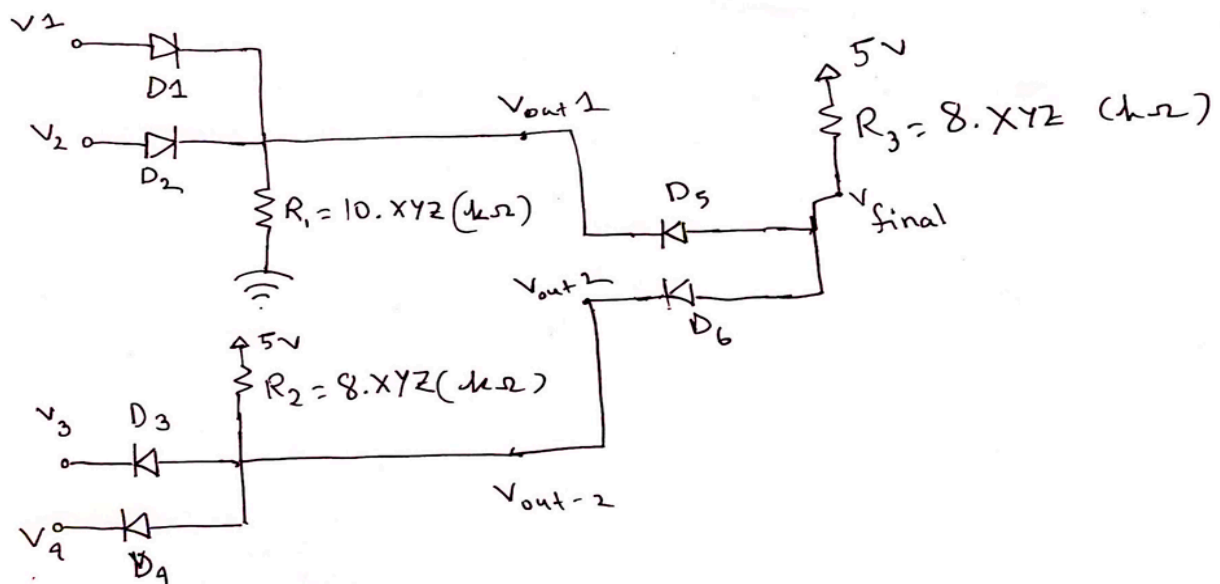
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- While posting your question here, please follow this format:

(a)	CO	Question	[Marks]
(b)	CO	Question	[Marks]
(c)	CO	Question	[Marks]

DL (OBR)



For the circuit shown above, there are total 4 input terminals marked as V_1 , V_2 , V_3 and V_4 . The input voltages can be either 5V (HIGH state) and 0V (LOW state). The junction voltage for each of the diode is 0.7 V. Here XYZ=last 3 digits of your student ID

(a)	CO1	Find the power dissipated by all three resistors when $V_1, V_3 = 5V$ and $V_2, V_4 = 0V$.	[12]
(b)	CO1	Find the input voltage combination for V_1, V_2, V_3 and V_4 where D_5 diode turns on and D_6 diode turns off. (you can write the answer in one line, for example : $V_1 = 5V, V_2 = 5V, V_3 = 5V$ and $V_4 = 5V$.)	[3]
(c)	CO1	Determine the boolean expression of V_{final} in terms of input V_1, V_2, V_3 and V_4	[5]

DL Solution:

① Suppose, $R_1 = 10k\Omega$, $R_2 = 8k\Omega$, $R_3 = 8k\Omega$

for $V_1 = 5V$ and $V_2 = 0V$, D_1 diode $\rightarrow$ turns on
 D_2 diode $\rightarrow$ turns off

$\therefore$ voltage at $V_{out1} = 4.3V$.
 $\therefore$ current through $R_1 = \frac{4.3}{10} mA = 0.43mA$.

$\therefore$ Power $= I_1^2 R_1 = (0.43)^2 \times 10 = \boxed{1.849 mW}$. [4marks]

for $V_3 = 5V$ and $V_4 = 0V$, D_3 diode $\rightarrow$ turns off
 D_4 " $\rightarrow$ turns on.

$\therefore$ voltage at $V_{out2} = 0.7V$.
 $\therefore$ current through $R_2 = \left(\frac{5 - 0.7}{8} \right) = 0.5375mA$
 $\therefore$ power $P_2 = I_2^2 R_2 = \boxed{2.31125 mW}$. [4marks]

for $V_{out1} = 4.3V$ and $V_{out2} = 0.7V$,

D_5 diode $\rightarrow$ turns off
 D_6 " $\rightarrow$ turns on.

$\therefore$ voltage at $V_{final} = 0.7 + 0.7 = 1.4V$. $\rightarrow$ [2marks]
 $\therefore$ current through $R_3 = \left(\frac{5 - 1.4}{8} \right) = 0.45mA$

$$\therefore \text{power} = (0.45)^2 \times 8 = \boxed{1.62 \text{ mW}} \rightarrow [2 \text{ marks}]$$

$$\therefore \text{total} = 1.62 + 2.3125 + 1.849 = \boxed{5.5825 \text{ mW}}$$

(b) $V_1 = 0V, V_2 = 0V, V_3 = 5V, V_4 = 5V$
[3 marks]

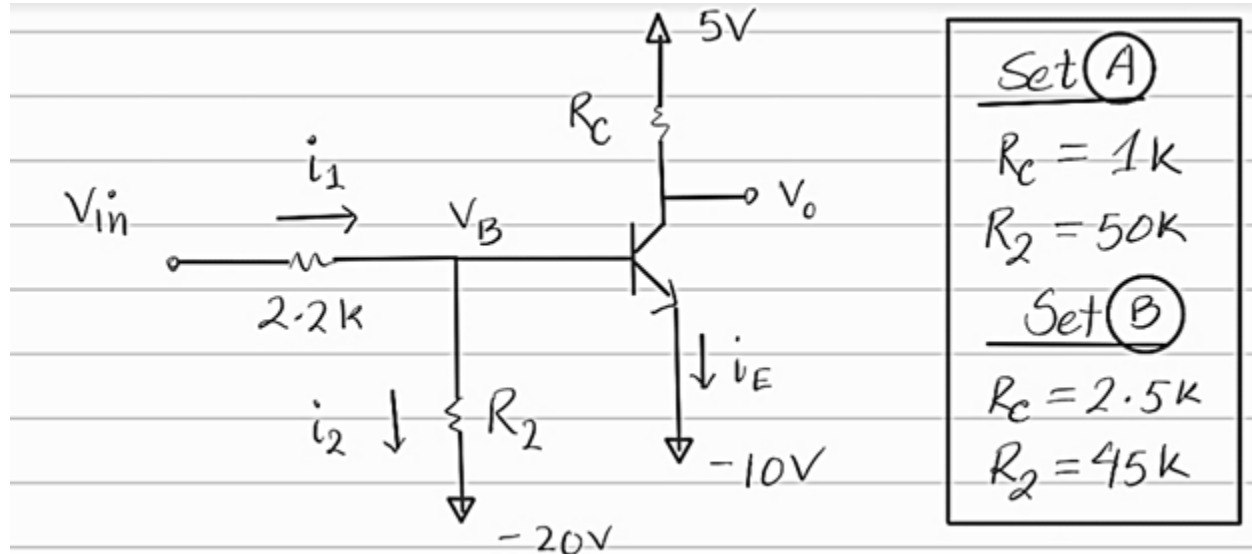
(c) $V_{\text{out } 1} = V_1 + V_2$ [boolean OR] [2 marks]

$V_{\text{out } 2} = V_3 \cdot V_4$ [boolean AND] [2 marks]

$V_{\text{final}} = V_{\text{out } 1} \cdot V_{\text{out } 2}$ [boolean AND] [1 mark]

$$\boxed{= (V_1 + V_2) \cdot V_3 \cdot V_4} \quad (\text{Ans})$$

RTL (MJA)



An RTL circuit is given above. β_F for the used transistor is known to be 45. The input voltage, v_{in} of the circuit can be in either of these two states:

- HIGH : 5V
- LOW : -10V

(a)	CO1	Find the voltages v_B and v_o for each input state.	[08]
(b)	CO1	Find the currents i_1 , i_2 , i_E when v_{in} is in HIGH state.	[06]
(c)	CO1	If an identical RTL load circuit is connected to the above circuits output, find the new current i_E of the driver. [v_{in} is in HIGH state] <u>Hint:</u> $i_E = i_B + i_C = i_B + (i_C' - i_L')$	[06]

(a)	CO1	Find the voltages v_B and v_o for each input combination.	[08]
(b)	CO1	If an identical RTL load circuit is connected to the above circuits output, find the current i_E of the driver. [here, v_{in} is in HIGH state] <u>Hint:</u> $i_E = i_B + i_C = i_B + (i_C' - i_L')$	[12]

RTL Solution:

(a) Find V_B & V_O for all input combinations

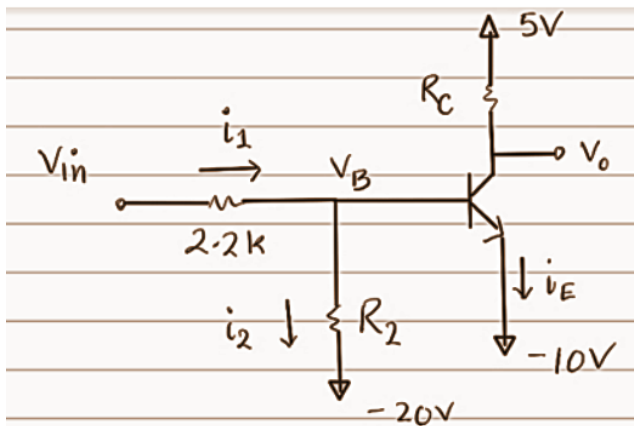
$$V_{in}(\text{low}) : \frac{V_B - (-10)}{2.2} + \frac{V_B - (-20)}{R_2} = 0$$

$$\Rightarrow V_B = -10.421 \text{ (A)} \mid -10.466 \text{ V (B)}$$

$$\text{So, } V_O = 5V$$

$$V_{in}(\text{high}) : V_B = -10 + 0.8 = -9.2V \text{ (A/B)}$$

$$V_O = -10 + 0.2 = -9.8V \text{ (A/B)}$$



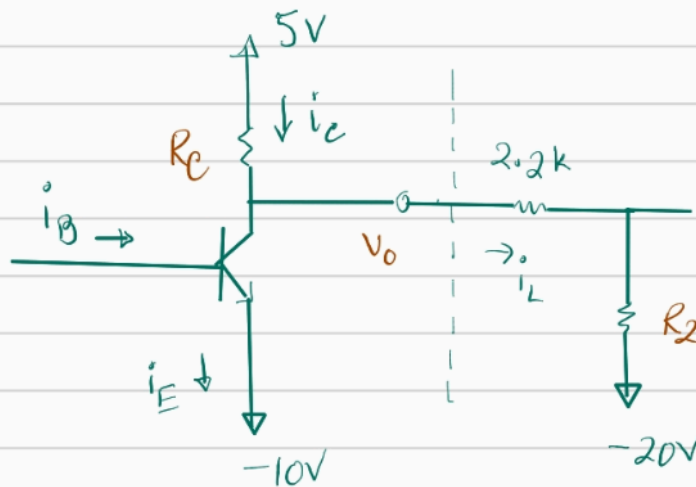
(b) Find i_1, i_2, i_E when $V_{in} = \text{high}$

$$V_{in}(\text{high}) : i_1 = \frac{5 - (-9.2)}{2.2} = 6.455 \text{ mA (A/B)}$$

$$i_2 = \frac{-9.2 - (-20)}{R_2} = 0.216 \mid 0.24 \text{ mA (A) (B)}$$

$$i_E = i_1 - i_2 = 6.239 \mid 6.215 \text{ mA (A) (B)}$$

If an identical load circuit is connected to this ckt, what would be i_E ?



Set (A)

$$R_C = 1k$$

$$R_2 = 50k$$

Set (B)

$$R_C = 2.5k$$

$$R_2 = 45k$$

$$i_E = i_B + i_C' = i_B + (i_C - i_L)$$

(A)

$$i_B = 6.239 \text{ mA}$$

$$i_L = \frac{-9.8 + 20}{2.2 + R_2}$$

$$= 0.195 \text{ mA}$$

$$i_C = 14.8 \text{ mA}$$

$$\therefore i_E = 6.239 + 14.8 - 0.195$$

$$= 20.844 \text{ mA}$$

(B)

$$i_B = 6.215 \text{ mA}$$

$$i_L = \frac{-9.8 + 20}{2.2 + R_2}$$

$$= 0.216 \text{ mA}$$

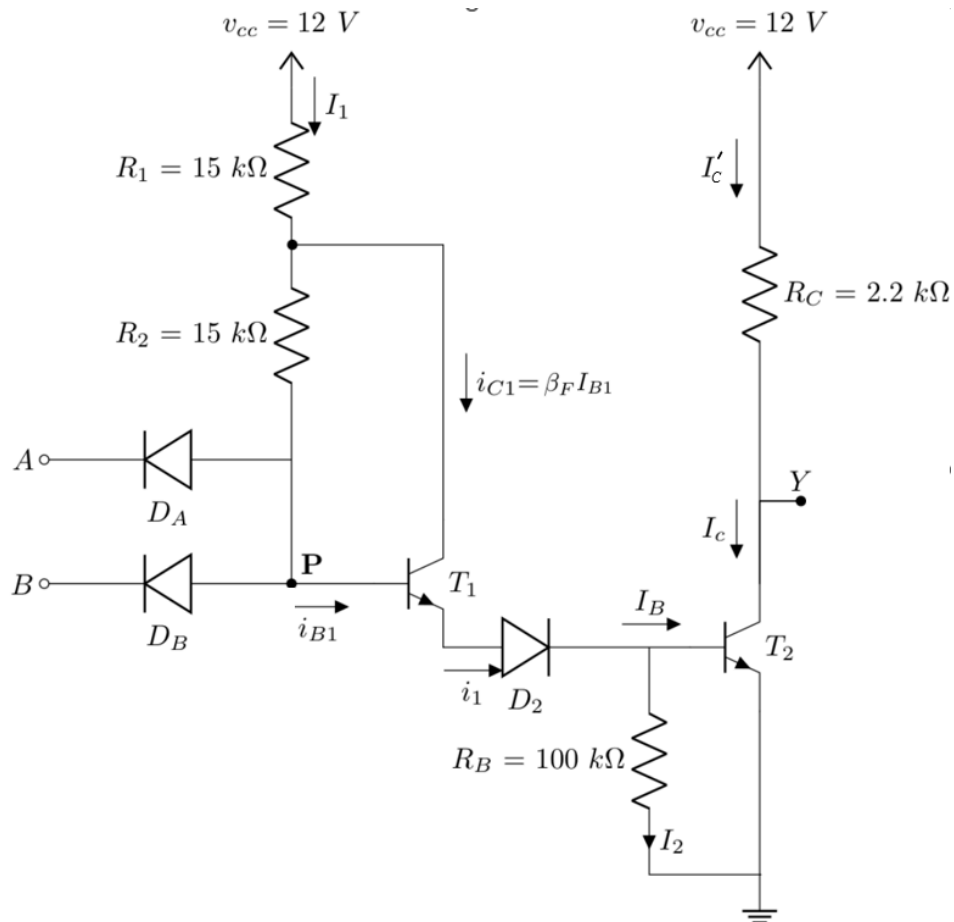
$$i_C = 5.92 \text{ mA}$$

$$\therefore i_E = 6.215 + 5.92 - 0.216$$

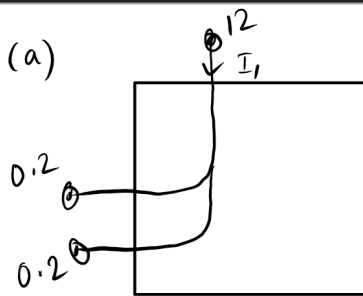
$$= 11.919 \text{ mA}$$

DTL (QSH)

For the DTL circuit in the figure, assume $\beta_F = 60$, $V_{BE}(\text{saturation}) = 0.8\text{V}$, and $V_{CE}(\text{saturation}) = 0.2\text{V}$. The input voltages 'A' and 'B' can be either 0V or 12V.



(a)	CO2	Determine the power dissipation for the DTL NAND gate when both the inputs 'A' and 'B' are LOW.	[4]
(b)	CO2	Identify what should be the minimum current gain, $\beta_{\min}$ to prevent the transistor from malfunctioning.	[7]
(c)	CO2	Estimate the maximum fanout this logic device can handle without any issues.	[9]



$$I_1 = \frac{12 - 0.9}{30k}$$

$$= 0.37 \text{ mA}$$

$\therefore$ Power Dissipation,

$$P = \Delta V I$$

$$= (12 - 0.2) \times 0.37$$

$$= 4.366 \text{ mW}$$

(b) $I_1 = I_{C1} + I_{B1}$

$$\Rightarrow I_1 = (\beta + 1) I_{B1}$$

$$\Rightarrow \frac{12 - V_{C1}}{15k} = (60 + 1) \frac{V_{C1} - 2.2}{15k}$$

$$\Rightarrow V_{C1} = 2.358 \text{ V}$$

$$I_{E1} = I_{C1} + I_{B1}$$

$$= I_1$$

$$= \frac{12 - 2.358}{15k}$$

$$\therefore I_{E1} = 0.643 \text{ mA}$$

$$I_{B2} = I_{E1} - I_2$$

$$= 0.643 - \frac{0.8}{100k}$$

$$= 0.635 \text{ mA}$$

Now, $I_{C2} = \frac{12 - 0.2}{2.2k}$

$$= 5.364 \text{ mA}$$

$$\therefore \beta_{\min} = \frac{I_{C2}}{I_{B2}}$$

$$= \frac{5.364}{0.635}$$

$$= 8.447$$

(c) For maximum Fanout,

Case 1 : Driver output HIGH

D_A and D_B are OFF

$$\therefore I_L = 0$$

$$\therefore \text{Fanout} = \frac{I_S}{I_L} = \infty$$

Case 2 : Driver output LOW,

From (a), $I_L = 0.37 \text{ mA}$

From (b), $I_{C'} = 5.364 \text{ mA}$

Now, $\beta_F \times I_{B2} = I_{C'} + N I_L$

$$\text{or, } 60 \times 0.635 = 5.364 + N \times 0.37$$

$$\text{or, } N = \lfloor 88.476 \rfloor$$

$$= 88$$

$$\therefore \text{Maximum Fanout} = \min(\infty, 88)$$

$$= 88$$

TTL

(a) (i) Driver o/p low $\rightarrow$ both i/p's high.
 $Q_1 \rightarrow$ Rev. active

$$I_1 = \frac{6-2.3}{12k} = 0.308 \text{ mA} \quad [\beta_R = 0.1]$$

$$I_{EX} = I_{EY} = \beta_R I_1 = 0.0308 \text{ mA}$$

$$I_{B2} = I_1 + 2I_{EX} = 0.3696 \text{ mA}$$

$$I_2 = \frac{6-0.9}{4k} = 1.275 \text{ mA}$$

$$I_4 = \frac{0.8}{2k} = 0.4 \text{ mA}$$

$$I_{B0} = I_2 + I_{B2} - I_4 = 1.2446 \text{ mA}$$

$$I_3 = \frac{6-0.1}{5} = 1.18 \text{ mA}$$

$$i_L = \frac{6-0.9}{12k} = 0.425 \text{ mA}$$

Driver o/p low

$$\downarrow$$

$$V_o = 0.1 \text{ V}$$

$G_1, G_2, \dots, G_N$

$\downarrow$
Saturation

$$I_c = I_3 + N \times i_L$$

$$\frac{I_c}{I_{B0}} < \beta_F$$

$$\rightarrow \frac{1.18 + N \times 0.425}{1.2446} < 30$$

$$N = 85$$

(ii) ^{load}
 if both i/p's $\rightarrow$ low
 $i_L \rightarrow i_L/2$ (halved)

$$N \times i_L < \beta_F \times I_{B0} - I_3$$

$i_L \downarrow \quad N \uparrow$ Fanout will increase.

$$(b) P = (6-0) \times (I_1 + I_2 + I_3 + I_{EX} + I_{EY}) + (0.1-0) \times N \times i_L$$

$$= 20.5601 \text{ mW}$$